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Fixing device and image forming apparatus

Abstract

A fixing device includes a fixing belt, a heat source, a fixing pad, and a support member. In the support member, a fixture and an upright are integrally formed, and the fixing pad is fixed in a rotation axis direction of the fixing belt. The upright is upright from a side opposite to the fixing pad in a width direction orthogonal to the rotation axis direction of the fixture, and a plurality of connections between the fixture and the upright is provided.

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Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2007/0189818	12/2006	Kondo	399/329	G03G 15/2064
2008/0298862	12/2007	Shinshi	399/324	G03G 15/2064
2010/0092220	12/2009	Hasegawa	399/328	G03G 15/2064
2010/0202809	12/2009	Shinshi	399/329	G03G 15/2064
2010/0290822	12/2009	Hasegawa	399/329	G03G 15/2064
2011/0026988	12/2010	Yoshikawa	399/329	G03G 15/2053
2011/0052282	12/2010	Shinshi	399/329	G03G 15/2064
2011/0064437	12/2010	Yamashina	399/329	G03G 15/205
2011/0217094	12/2010	Hasegawa	399/329	G03G 15/20
2011/0229200	12/2010	Yamaguchi	399/122	G03G 15/2039
2012/0008971	12/2011	Lee	399/328	G03G 15/2064
2012/0308254	12/2011	Suzuki	399/329	G03G 15/2053
2013/0084111	12/2012	Iwata	399/329	G03G 15/2053
2016/0011548	12/2015	Utsunomiya	399/329	G03G 15/206
2016/0378027	12/2015	Sawada	399/329	G03G 15/2053
2016/0378028	12/2015	Fujiwara	399/323	G03G 15/2028
2017/0010570	12/2016	Fujimoto	N/A	G03G 15/2053
2017/0185009	12/2016	Yoshinaga	N/A	G03G 15/2042
2017/0255147	12/2016	Seki	N/A	G03G 15/2053
2017/0261900	12/2016	Sawada	N/A	G03G 15/206
2017/0364000	12/2016	Seki	N/A	G03G 15/2053
2018/0011432	12/2017	Yoshinaga	N/A	G03G 15/2042
2018/0017910	12/2017	Sawada	N/A	G03G 15/2053
2018/0157198	12/2017	Seto	N/A	G03G 15/2025
2018/0173138	12/2017	Seki	N/A	G03G 15/2017
2018/0284663	12/2017	Matsuda	N/A	G03G 15/2042
2018/0284669	12/2017	Fujimoto	N/A	G03G 15/2053
2023/0251594	12/2022	Yoshikawa	399/328	G03G 15/2064

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2011-197019	12/2010	JP	N/A
2024148049	12/2023	JP	N/A

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) The present application claims priority from Japanese Application JP2023-061368, the content to which is hereby incorporated by reference into this application.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(2) The present disclosure relates to a fixing device and an image forming apparatus such as a copier, a multifunction peripheral, a printer, or a facsimile machine.

2. Description of the Related Art

(3) FIG. 8 is a front view illustrating an example of a main part of a conventional fixing device 12X. FIGS. 9A and 9B are an exploded perspective view and a bottom view, respectively, illustrating an example of a support member 33X in the conventional fixing device 12X.

(4) As illustrated in FIG. 8, the conventional fixing device 12X includes a cylindrical fixing belt 31, a heat source 36 that heats the fixing belt 31 inside the fixing belt 31, a fixing pad 34 that is in sliding contact with an inner surface 31a of the fixing belt 31, and a support member 33X that supports the fixing pad 34. In FIG. 8, the reference numeral 32 denotes a pressure roller.

(5) Specifically, the support member 33X includes a first support 33X1 and a second support 33X2. The first support 33X1 and the second support 33X2 are formed into an L-shape in a front view (see FIG. 8). In the support member 33X, respective outer surfaces of the L-shaped first support 33X1 and the L-shaped second support 33X2 are overlapped with each other to form a T-shape in a front view.

(6) In the conventional fixing device 12X described above, since the two supports are overlapped with each other, a large amount of heat transfers from the fixing belt 31 to the support member 33X through the fixing pad 34. Specifically, in the support member 33X, uprights 332X, which are upright from fixtures 331X to fix the fixing pad 34, are overlapped with each other, and the volume of the support member 33X is large, and therefore heat easily shifts from the fixtures 331X to the uprights 332X. As described above, when a large amount of heat transfers from the fixing belt 31 to the support member 33X through the fixing pad 34, the heat energy transmitted from the heat source 36 to the fixing belt 31 increases, and the power consumption increases accordingly.

(7) In this regard, there is a conventionally known configuration in which a reinforcement member (support member) includes a pair of reinforcement portions (uprights) that are upright from both one end and the other end in a width direction orthogonal to a rotation axis direction of a fixing belt on the side opposite to a nip forming member (fixing pad) from a reinforcement recess (fixture).

SUMMARY OF THE INVENTION

(8) However, in the conventional configuration, the configuration of the reinforcement member (support member) is complicated.

(9) Thus, an object of the present disclosure is to provide a fixing device that includes a cylindrical fixing belt, a heat source that heats the fixing belt inside the fixing belt, a fixing pad that is in sliding contact with an inner surface of the fixing belt, and a support member that supports the fixing pad and, that may reduce the volume of the support member and thus suppress heat transfer from the fixing belt to the support member through the fixing pad, and an image forming apparatus.

(10) A fixing device according to an aspect of the present disclosure includes: a cylindrical fixing belt; a heat source that heats the fixing belt inside the fixing belt; a fixing pad that is in sliding contact with an inner surface of the fixing belt; and a support member that supports the fixing pad, wherein the support member includes a fixture that fixes the fixing pad in a rotation axis direction of the fixing belt and an upright that is upright from a side opposite to the fixing pad between one end of the fixture on one side and the other end on the other side in a width direction orthogonal to

the rotation axis direction, the fixture and the upright being integrally formed, and includes a plurality of connections between the fixture and the upright.

(11) Further, an image forming apparatus according to the present disclosure includes the fixing device according to the present disclosure.

(12) According to the present disclosure, it is possible to reduce the volume of the support member and thus suppress heat transfer from the fixing belt to the support member through the fixing pad.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a cross-sectional view illustrating a schematic configuration of an image forming apparatus including a fixing device according to the present embodiment.

(2) FIG. 2A is a perspective view of the fixing device in the image forming apparatus illustrated in FIG. 1.

(3) FIG. 2B is a cross-sectional view taken along the line A-A illustrated in FIG. 2A.

(4) FIG. 3A is a perspective view of an end part on one side in a rotation axis direction of a fixing belt of the fixing device according to the present embodiment as viewed from the front side.

(5) FIG. 3B is a perspective view of an end part on the other side in the rotation axis direction of the fixing belt of the fixing device according to the present embodiment as viewed from the back side.

(6) FIG. 4 is a front view illustrating an example of a main part of the fixing device.

(7) FIG. 5A is a perspective view illustrating an example of a support member in the fixing device.

(8) FIG. 5B is a bottom view illustrating an example of the support member in the fixing device.

(9) FIG. 6 is a perspective view illustrating another example of the support member in the fixing device.

(10) FIG. 7 is a perspective view illustrating still another example of the support member in the fixing device.

(11) FIG. 8 is a front view illustrating an example of a main part of a conventional fixing device.

(12) FIG. 9A is an exploded perspective view illustrating an example of a support member in the conventional fixing device.

(13) FIG. 9B is a bottom view illustrating an example of the support member in the conventional fixing device.

DETAILED DESCRIPTION OF THE INVENTION

(14) Hereinafter, aspects of the present disclosure will be described with reference to the drawings. In the following description, the same components are denoted by the same reference numerals. Their names and functions are also the same. Therefore, detailed description thereof will not be repeated.

Image Forming Apparatus

(15) FIG. 1 is a cross-sectional view illustrating a schematic configuration of an image forming apparatus **100** including a fixing device **12** according to an aspect of the present disclosure.

(16) As illustrated in FIG. 1, the image forming apparatus **100** is a multifunction peripheral having a copy function, a scanner function, a facsimile function, and a printer function, and transmits an image of a document G read by an image reading device **102** to the outside. Further, the image forming apparatus **100** forms an image of the document G read by the image reading device **102** or an image received from outside on a sheet P such as a recording sheet in color or monochrome. The image forming apparatus **100** may be a monochrome image forming apparatus. Further, the image forming apparatus **100** may be a color image forming apparatus of other forms.

(17) A document feeding device **160** (automatic document conveying device) supported so as to be openable and closable with respect to an image reader **130** is provided on the upper side of the

image reader **130**. The image reading device **102** reads the document G conveyed by the document feeding device **160**. The document feeding device **160** includes a document placement tray **161** on which the document G is placed and a document discharge tray **162** on which the discharged document G is stacked. The document feeding device **160** sequentially conveys the one or more documents G placed on the document placement tray **161** one by one onto a document reader **130b** in the image reader **130** and discharges the documents G to the document discharge tray **162**. Further, the image reader **130** includes a document placement table **130a** on which the document G is placed. The image reading device **102** reads the document G placed on the document placement table **130a**. In the image forming apparatus **100**, when the document feeding device **160** is opened, the document placement table **130a** above the image reader **130** is opened, and the document G may be placed by hand. The image reader **130** reads the document G conveyed by the document feeding device **160** in a state where a scanning optical system **130c** is positioned at a reading position below the document reader **130b** or reads the document G placed on the document placement table **130a** by scanning the scanning optical system **130c** to generate image data.

(18) An image forming apparatus main body **101** includes an optical scanning device **1**, a developing device **2**, a photosensitive drum **3**, a drum cleaning device **4**, a charger **5**, an intermediate transfer belt device **70**, a secondary transfer device **11**, a fixing device **12**, a sheet conveyance path S, a sheet feed cassette **18**, and a sheet discharge tray **141**.

(19) The image forming apparatus **100** handles image data corresponding to a color image using each color of black (K), cyan (C), magenta (M), and yellow (Y) or a monochrome image using a single color (for example, black). An image former **50** of the image forming apparatus **100** includes the four developing devices **2**, the four photosensitive drums **3**, the four drum cleaning devices **4**, and the four chargers **5** to form four types of toner images, which correspond to black, cyan, magenta, and yellow, respectively, and constitute four image stations Pa, Pb, Pc, and Pd.

(20) The charger **5** uniformly charges the surface of the photosensitive drum **3** to a predetermined potential. The optical scanning device **1** forms an electrostatic latent image by exposing the surface of the photosensitive drum **3**. The developing device **2** develops the electrostatic latent image on the surface of the photosensitive drum **3** to form a toner image on the surface of the photosensitive drum **3**. The drum cleaning device **4** removes and collects residual toner on the surface of the photosensitive drum **3**. By the series of operations described above, a toner image of each color is formed on the surface of each of the photosensitive drums **3**.

(21) The intermediate transfer belt device **70** includes an intermediate transfer roller **6**, an endless intermediate transfer belt **71**, an intermediate transfer driving roller **72**, an intermediate transfer driven roller **73**, and a cleaning device **9**. The four intermediate transfer rollers **6** are provided inside the intermediate transfer belt **71** so as to form four types of toner images corresponding to the respective colors. The intermediate transfer roller **6** transfers the toner image of each color formed on the surface of the photosensitive drum **3** to the intermediate transfer belt **71** circulating in a circulating direction C.

(22) The intermediate transfer belt **71** is stretched around the intermediate transfer driving roller **72** and the intermediate transfer driven roller **73**. In the image forming apparatus **100**, the toner images of the respective colors formed on the surfaces of the respective photosensitive drums **3** are sequentially transferred and superimposed on the surface of the intermediate transfer belt **71** to form a color toner image on the surface of the intermediate transfer belt **71**.

(23) A secondary transfer device **11** forms a transfer nip TN between a secondary transfer roller **11a** and the intermediate transfer belt **71** and conveys the sheet P conveyed through the sheet conveyance path S while sandwiching the sheet P in the transfer nip TN. When the sheet P passes through the transfer nip TN, the toner image on the surface of the intermediate transfer belt **71** is transferred to the sheet P by the secondary transfer device **11**, and the sheet P is conveyed to the fixing device **12**. The cleaning device **9** removes and collects waste toner remaining on the surface of the intermediate transfer belt **71** without being transferred to the sheet P.

(24) The fixing device **12** includes a fixing belt **31** and the pressure roller **32** that rotate with the sheet **P** interposed therebetween. The fixing device **12** nips the sheet **P** on which the toner image has been transferred between the fixing belt **31** and the pressure roller **32**, applies heat and pressure to the sheet **P**, and fixes the toner image to the sheet **P**. Although not illustrated in FIG. **1**, the fixing device **12** includes components other than the fixing belt **31** and the pressure roller **32**. Details of the fixing device **12** will be described below.

(25) The sheet feed cassette **18** is a cassette to store the sheets **P** used for image formation and is provided under the optical scanning device **1**. The sheet **P** is pulled out from the sheet feed cassette **18** by a pickup roller **16** and conveyed to the sheet conveyance path **S**. The sheet **P** conveyed to the sheet conveyance path **S** is conveyed to a discharge roller **17** via the secondary transfer device **11** and the fixing device **12** and is discharged to the sheet discharge tray **141** in a discharger **140**. A conveyance roller **13**, a registration roller **14**, and the discharge roller **17** are provided in the sheet conveyance path **S**. The conveyance roller **13** promotes conveyance of the sheet **P**. The registration roller **14** temporarily stops the sheet **P** and aligns the leading edge of the sheet **P**. The registration roller **14** conveys the temporarily stopped sheet **P** in accordance with the timing of the toner image on the intermediate transfer belt **71**.

(26) In FIG. **1**, the one sheet feed cassette **18** is provided, but the present disclosure is not limited thereto, and the plurality of sheet feed cassettes **18** may be provided, and the different types of sheets **P** may be stacked in the respective sheet feed cassettes **18**.

(27) Further, the image forming apparatus **100** conveys the sheet **P** from the discharge roller **17** to a sheet reversing path **Sr** in a reverse direction in a case where an image is formed not only on the front surface of the sheet **P** but also on the back surface thereof. The image forming apparatus **100** reverses the sheet **P** conveyed in the reverse direction, and guides the sheet **P** to the registration roller **14** again. Further, the image forming apparatus **100** forms an image on the back surface of the sheet **P** guided to the registration roller **14** in the same manner as the front surface, and discharges the sheet **P** to the sheet discharge tray **141**.

Fixing Device

(28) FIG. **2A** is a perspective view of the fixing device **12** in the image forming apparatus **100** illustrated in FIG. **1**. FIG. **2B** is a cross-sectional view taken along the line A-A illustrated in FIG. **2A**.

(29) The fixing device **12** further includes a support member **33**, a fixing pad **34**, a heat source **36**, a reflection member **37** (a reflection plate in this example), a temperature detector **38** (a thermopile in this example), a separation plate **39**, and a thermostat **40**. The support member **33**, the fixing pad **34**, the heat source **36**, and the reflection member **37** are provided inside the fixing belt **31**.

(30) The fixing belt **31** is a cylindrical [endless (annular)] flexible belt. The fixing belt **31** is rotatable around a rotation axis α along an orthogonal direction that is orthogonal to a conveying direction **H** of the sheet **P**.

(31) The fixing pad **34** is made of, for example, resin, and is formed into an elongated plate shape extending in a rotation axis direction **W** of the fixing belt **31**. The fixing pad **34** includes a fixing pad main body **340** and a sliding sheet **341** provided on a surface of the fixing pad main body **340** on a side in sliding contact with the fixing belt **31**. The fixing pad **34** is preferably made of a resin material having heat resistance, and for example, a heat-resistant resin material having heat resistance, such as liquid crystal polymer (LCP) or polyether ether ketone (PEEK), may be used.

(32) The support member **33** is a member that supports the fixing pad **34**. Both ends of the support member **33** in the rotation axis direction **W** are fixed to a fixing device main body **12a** (main body frame). The reflection member **37** is provided on at least a surface of the support member **33** on the heat source **36** side.

(33) The heat source **36** is a member that heats the fixing belt **31** and extends in the rotation axis direction **W** of the fixing belt **31**. The heat source **36** may be, for example, a lamp heater such as a halogen lamp. The fixing belt **31** is heated to a predetermined fixing temperature (for example,

160° C. to 250° C., in this example, 160° C.) by the heat source **36**. Therefore, not only the fixing belt **31** but also the sliding sheet **341** and the like have heat resistance to the above temperature.

(34) The reflection member **37** is provided to cover at least the surface of the support member **33** on the heat source **36** side and is formed of a thin plate-like metal member in this example. Thus, the fixing belt **31** may be efficiently heated. The reflection member **37** is fixed to the support member **33**.

(35) The pressure roller **32** is provided at a position opposed to the fixing pad **34** with the fixing belt **31** interposed therebetween. The pressure roller **32** rotates about a rotation axis parallel to the rotation axis α of the fixing belt **31** and extends parallel to the fixing belt **31**. The pressure roller **32** presses the fixing belt **31** toward the fixing pad **34** to form a fixing nip FN between the pressure roller **32** and the fixing belt **31**.

(36) A driving force from a drive source (not illustrated) such as a motor is transmitted to the pressure roller **32** via a drive transmission mechanism (not illustrated) such as a gear. The pressure roller **32** is rotationally driven by receiving the driving force from the driving source. The fixing belt **31** is driven to rotate in a first rotation direction R1 opposite to a second rotation direction R2 of the pressure roller **32** in accordance with the rotational driving of the pressure roller **32**. That is, the pressure roller **32** forms the fixing nip FN by coming into contact with the surface of the fixing belt **31** and transmits a driving force to the fixing belt **31** via the fixing nip FN to drive and rotate the fixing belt **31**.

(37) The temperature detector **38** detects the surface temperature of the fixing belt **31**. In the fixing device **12**, the temperature of the heat source **36** is controlled based on the temperature detected by the temperature detector **38** so that the fixing belt **31** has a fixing temperature (160° C. in this example). In this example, the temperature detector **38** is fixed to the image forming apparatus main body **101** (main body frame) (see FIG. 1). The temperature detector **38** detects the surface temperature of the fixing belt **31** in a non-contact manner.

(38) The separation plate **39** is provided in the vicinity of the downstream side of the fixing nip FN in the first rotation direction R1 of the fixing belt **31** and prevents the sheet P from winding around the fixing belt **31**.

(39) The thermostat **40** cuts off power supply to the heat source **36** when the heat source **36** is abnormally heated. Specifically, the thermostat **40** is electrically connected to a power line (not illustrated) for supplying power to the heat source **36** and directly cuts off the power supply to the heat source **36**. The thermostat **40** is fixed to the fixing device main body **12a** (main body frame). When a predetermined reaction temperature (operating temperature, rated temperature) is obtained, the thermostat **40** operates to cut off the power supply to the heat source **36**.

First Embodiment

(40) FIGS. 3A and 3B are perspective views of end parts on one side W1 and the other side W2 in the rotation axis direction W of the fixing belt **31** of the fixing device **12** according to the present embodiment as viewed from the front side and the back side, respectively. FIG. 4 is a front view illustrating an example of a main part of the fixing device **12**. FIGS. 5A and 5B are a perspective view and a bottom view illustrating an example of the support member **33** in the fixing device **12**, respectively.

(41) As illustrated in FIGS. 3A and 3B, the fixing device **12** includes a pair of holding members **411** and **412** (41) (belt guide members) and the fixing belt **31** provided on an outer peripheral surface **401a** of the pair of holding members **411** and **412** (41) so as to be rotatable about the rotation axis α (see FIG. 2B). The pair of holding members **411** and **412** (41) are fixed to the fixing device main body **12a** (main body frame) (see FIG. 2B).

(42) The pair of holding members **411** and **412** (41) hold both ends of the inner surface **31a** of the fixing belt **31** in the rotation axis direction W. Of the pair of holding members **411** and **412** (41), the one holding member **411** holds the end on the one side W1 (the front side, the operating side) in the rotation axis direction W of the inner surface **31a** of the fixing belt **31**, and the other holding

member **412** holds the end on the other side **W2** (the back side) in the rotation axis direction **W** of the inner surface **31a** of the fixing belt **31**.

(43) As illustrated from FIGS. **3A** to **5B**, the fixing device **12** includes the cylindrical fixing belt **31**, the heat source **36** that is provided inside the fixing belt **31** to heat the fixing belt **31**, the fixing pad **34** that is provided inside the fixing belt **31** and is in sliding contact with the inner surface **31a** of the fixing belt **31**, and the support member **33** that supports the fixing pad **34**. The fixing belt **31**, the heat source **36**, the fixing pad **34**, and the support member **33** are members elongated in the rotation axis direction **W**.

(44) In the support member **33**, a fixture **331** and an upright **332** are integrally formed. The fixture **331** fixes the fixing pad **34** in the rotation axis direction **W** of the fixing belt **31**. The upright **332** is upright from the side opposite to the fixing pad **34** between one end **33a** (see FIG. **4**) of the fixture **331** on one side **V1a** and the other end **33b** (see FIG. **4**) on the other side **V1b** in a width direction **V1** orthogonal to the rotation axis direction **W**. The upright **332** extends in the rotation axis direction **W**. A plurality of connections **333** (see FIGS. **5A** and **5B**) between the fixture **331** and the upright **332** is provided. Specifically, a space [one or a plurality of (in this example, a plurality of) openings (in this example, through-holes **331b** to **331b**)] is provided between the adjacent connections **333** to **333**.

(45) According to the present embodiment, in the support member **33**, the fixture **331** and the upright **332** are integrally formed, and the two supports are not overlapped with each other as in the configuration of the related art, and therefore, it is possible to reduce the volume of the support member **33** and thus suppress heat transfer from the fixing belt **31** to the support member **33** through the fixing pad **34**. Specifically, the upright **332** may be formed integrally (as a single piece) so that the volume of the upright **332** may be reduced, and the heat capacity of the support member **33** may be reduced accordingly. In addition, in the support member **33**, since the plurality of connections **333** to **333** between the fixture **331** and the upright **332** is provided, it is possible to shorten the entire lengths of the connections **333** to **333** in the rotation axis direction **W** by the space (**331b** to **331b**) provided between the adjacent connections **333** and **333**. Thus, it is possible to make it difficult for heat to shift from the fixture **331** to the upright **332**.

(46) Therefore, it is possible to suppress heat transfer from the fixing belt **31** to the support member **33** through the fixing pad **34**. As a result, the thermal energy transmitted from the heat source **36** to the fixing belt **31** may be reduced, and the power consumption may be reduced accordingly.

(47) Further, since the fixture **331** and the upright **332** are integrally formed, the configuration of the support member **33** may be simplified.

(48) Moreover, since the support member **33** is upright between the one end **33a** and the other end **33b** of the fixture **331** in the width direction **V1**, the fixture **331** pressed by the pressure roller **32** may be supported by the support member **33**, for example, as compared with a case where the support member **33** is upright (cantilevered) from the one end **33a** or the other end **33b** of the fixture **331** in the width direction **V1**.

(49) According to the present embodiment, in the support member **33**, the fixture **331** and the upright **332** are formed into a T-shape in a front view by bending a single plate (sheet metal) (by bending process). Specifically, the connections **333** to **333** are bends. The upright **332** is formed at a right angle or a substantially right angle with respect to the fixture **331**. A width **d2** (see FIG. **4**) (height) of the upright **332** in a width direction **V2** orthogonal to the rotation axis direction **W** is equal to or larger than (in this example, larger than) a width **d1** (see FIG. **4**) of the fixture **331** in the width direction **V1** orthogonal to the rotation axis direction **W**.

(50) With this configuration, since the support member **33** is formed by bending a single plate, it is possible to reduce the number of parts, and thus improve the part accuracy of the support member **33**, and in addition, it is possible to reduce the volume of the support member **33**, and thus suppress heat transfer from the fixing belt **31** to the support member **33** through the fixing pad **34**. In addition, it is possible to secure a space inside the fixing belt **31**, or it is possible to reduce the

diameter of the fixing belt **31** and thus reduce the size of the fixing device.

(51) According to the present embodiment, an opposed surface **331a** (see FIGS. **4** and **5B**) of the fixture **331** opposed to the fixing pad **34** is formed into a planar shape.

(52) With this configuration, the fixture **331** may be fixed to the fixing pad **34** with the planar opposed surface **331a** provided on the fixture **331** itself.

(53) According to the present embodiment, the upright **332** includes through-holes **331b** to **331b** adjacent to the connections **333** to **333**, and the fixture **331** includes a fixture main body **3311** (see FIGS. **5A** and **5B**) closer to the one side **V1a** than the upright **332** in the width direction **V1** and protrusions **3312** to **3312** that protrude from the end on the other side **V1b** in the width direction **V1** toward the other side **V1b** beyond the upright **332** at the positions corresponding to the through-holes **331b** to **331b** (see FIGS. **5A** and **5B**) in the fixture main body **3311**. Specifically, the protrusions **3312** to **3312** are parts of the portions that are not bent as the upright **332** in the single plate forming the support member **33** and remains in the fixture **331**.

(54) With this configuration, the volume of the support member **33** may be reduced by the through-holes **331b** to **331b**, and the heat capacity of the support member **33** may be reduced accordingly. Thus, it is possible to suppress heat transfer from the fixing belt **31** to the support member **33** through the fixing pad **34**. Therefore, the thermal energy transmitted from the heat source **36** to the fixing belt **31** may be reduced, and the power consumption may be reduced accordingly.

(55) Specifically, in the individual through-holes **331b** to **331b**, the one protrusion **3312** is formed for the one through-hole **331b**. One end and the other end of the protrusion **3312** in the rotation axis direction **W** are respectively positioned near one end and the other end of the through-hole **331b** in the rotation axis direction **W**. In this way, the protrusions **3312** to **3312** may be easily formed, and the support member **33** may be easily manufactured accordingly.

(56) In the fixture **331**, a length **d1a** (see FIG. **4**) of the protrusions **3312** to **3312** in the width direction **V1** is smaller than a length **d1b** (see FIG. **4**) of the fixture main body **3311** in the width direction **V1**. Thus, it is possible to easily bend the fixture main body **3311**, and accordingly, it is possible to easily manufacture the support member **33**.

Second Embodiment

(57) FIG. **6** is a perspective view illustrating another example of the support member **33** in the fixing device **12**.

(58) According to the second embodiment, substantially the same components as those of the first embodiment are denoted by the same reference numerals, and the description thereof will be omitted.

(59) In the support member **33** according to the second embodiment, the numbers of the protrusions **3312** to **3312** and the through-holes **331b** to **331b** in the support member **33** according to the first embodiment are reduced, and the lengths of the protrusions **3312** to **3312** and the through-holes **331b** to **331b** in the rotation axis direction **W** are increased. One end and the other end of the protrusion **3312** in the rotation axis direction **W** are respectively positioned near one end and the other end of the through-hole **331b** in the rotation axis direction **W**.

(60) With this configuration, the volume of the support member **33** may be further reduced by the through-holes **331b** to **331b**, and the heat capacity of the support member **33** may be reduced accordingly. Thus, the heat transfer from the fixing belt **31** to the support member **33** through the fixing pad **34** may be further suppressed. Therefore, the thermal energy transmitted from the heat source **36** to the fixing belt **31** may be further reduced, and the power consumption may be reduced accordingly.

Third Embodiment

(61) FIG. **7** is a perspective view illustrating still another example of the support member **33** in the fixing device **12**.

(62) According to the present embodiment, an interposition member **35** (see FIGS. **4** and **7**) interposed between the fixture **331** of the support member **33** and the fixing pad **34** is further

provided. The interposition member **35** is a pad fixing member (sheet metal) for fixing the fixing pad **34** and is a member elongated in the rotation axis direction **W**.

(63) The fixing pad **34** may warp in the width direction **V1**, and in this case, the interposition member **35** may warp along with the warp of the fixing pad **34**. Then, a gap is formed between the interposition member **35** and the fixture **331** of the support member **33**. In this case, it is difficult to support the fixing pad **34** by the support member **33** through the interposition member **35**.

(64) In this regard, a plurality of projections **35a** to **35a**, which projects to the fixture **331** side and comes into contact with the fixture **331**, is provided on the interposition member **35** on the fixture **331** side. The protrusions **3312** to **3312** are provided at positions **Q** to **Q** corresponding to the projections **35a** to **35a** in the fixture main body **3311**. Specifically, the plurality of (two in this example) projections **35a** to **35a** is arranged in a row in the width direction **V1** at each of the positions **Q** to **Q**. Thus, the fixing pad **34** may be supported by the support member **33** through the interposition member **35**.

(65) In addition, since the interposition member **35** includes the plurality of projections **35a** to **35a**, the heat transfer from the fixing pad **34** to the support member **33** may be suppressed, and therefore, the heat energy transmitted from the heat source **36** to the fixing belt **31** may be reduced, and the power consumption may be reduced accordingly.

(66) As the protrusions **3312** to **3312** are provided at the positions **Q** to **Q** corresponding to the projections **35a** to **35a** in this manner, the positions where the projections **35a** to **35a** of the interposition member **35** are in contact with the fixture **331** may be ensured in the protrusions **3312** to **3312**.

(67) Specifically, in the individual through-holes **331b** to **331b**, the plurality of (two in this example) protrusions **3312**, **3312** is formed for the one through-hole **331b**. The plurality of protrusions **3312** and **3312** in each of the through-holes **331b** to **331b** is provided at intervals in the rotation axis direction **W**. In this way, the volume of the support member **33** may be further reduced by the through-holes **331b** to **331b**, and the heat capacity of the support member **33** may be further reduced accordingly.

(68) The present disclosure is not limited to the embodiments described above and may be implemented in various other forms. Therefore, the embodiments are merely examples in all respects, and should not be interpreted in a limited manner. The scope of the present disclosure is defined by the appended claims and is not limited by the description of the specification. Furthermore, all modifications and changes belonging to the equivalent scope of the claims are within the scope of the present disclosure.

Claims

1. A fixing device comprising: a cylindrical fixing belt; a heat source, that heats the cylindrical fixing belt, located inside the cylindrical fixing belt; a fixing pad that is in sliding contact with an inner surface of the cylindrical fixing belt; and a support member that supports the fixing pad, wherein the support member includes a fixture that fixes the fixing pad in a rotational axis direction of the cylindrical fixing belt and an upright that is upright from a side opposite the fixing pad between one end of the fixture on one side and the other end of the fixture on the other side in a width direction orthogonal to the rotational axis direction, the fixture and the upright being integrally formed, the support member further includes a plurality of connections between the fixture and the upright, in the support member, the fixture and the upright are formed into a T-shape in a front view by bending a single plate, an opposed surface of the fixture opposite the fixing pad is formed into a planar shape, the upright includes a through-hole between adjacent ones of the plurality of connections, and the fixture includes a fixture main body closer to the one side than to the upright in the width direction, and a protrusion that protrudes from an end on the other side in the width direction toward a side beyond the upright, the protrusion being provided, in the fixture

main body, at a position corresponding to the through-hole provided in the upright, the protrusion is a part of a portion that is not bent as the upright in the single plate of the support member and that remains in the fixture, and one end and the other end of the protrusion in the rotational axis direction are, respectively, positioned near one end and the other end of the through-hole in the rotational axis direction.

2. The fixing device according to claim 1, further comprising an interposition member interposed between the fixture of the support member and the fixing pad, wherein a plurality of projections that projects toward the fixture is provided on the interposition member on a fixture side, and the protrusion is further provided at a position corresponding to the plurality of projections provided on the interposition member, the interposition member being a pad fixing member for fixing the fixing pad.

3. An image forming apparatus comprising the fixing device according to claim 1.
